

Entity Relationship Analysis Report

Project Title: E-Commerce Order Management Database System

Phase: Week 2 - Relationship & Cardinality Analysis

1. Purpose & Scope

This report documents the structural relationships, operational constraints, and cardinality interactions between the business entities defined for the E-Commerce Order Management Database System. It ensures referential integrity and maps structural requirements to logical database dependencies.

2. Relationship & Cardinality Matrix

The matrix below serves as a definitive roadmap mapping out how database entities interact structurally with one another.

Source Entity	Target Entity	Relationship Description	Cardinality	Foreign Key Location
Customer	Order	A customer places an order	1 : N (One-to-Many)	Order (CustomerID)
Category	Product	A category classifies products	1 : N (One-to-Many)	Product (CategoryID)
Order	OrderItem	An order contains line items	1 : N (One-to-Many)	OrderItem (OrderID)
Product	OrderItem	A product is featured in line items	1 : N (One-to-Many)	OrderItem (ProductID)
Order	Payment	An order triggers a payment transaction	1 : 1 (One-to-One)	Payment (OrderID) [UNIQUE]

3. Detailed Structural Dependencies

Customer ↔ Order

- Business Rule:** A registered customer can execute zero, one, or many distinct orders over time. Every single order record must register back to a single unique, verified customer account.

- **Participation:** Optional for Customer (a user can register and buy nothing yet); Mandatory for Order (an order cannot exist without a valid purchasing customer).

Category ↔ Product

- **Business Rule:** To maintain stock organization, a category acts as a structural folder containing multiple products. Each product belongs strictly to a single category.
- **Participation:** Optional for Category (a new category can exist before items are assigned); Mandatory for Product.

Order ↔ Product (Resolved via OrderItem)

- **Business Rule:** Orders and Products share an implicit Many-to-Many conceptual dependency. To preserve clean schema standardizations, the associative bridge entity `OrderItem` acts as the intermediary.
- **Participation:** An `OrderItem` record safely depends on the mandatory existences of both an active `OrderID` and an active `ProductID`.

Order ↔ Payment

- **Business Rule:** Every processed order has exactly one payment cycle associated with it to prevent duplicate billings or orphaned collections.
- **Participation:** Mandatory for Payment (a payment must attach directly to a real invoice order record). The `OrderID` within the `Payment` entity acts as a Foreign Key equipped with a strict `UNIQUE` constraint to solidify the 1:1 bond.

4. Referential Integrity Rules

- **ON UPDATE CASCADE:** If an underlying primary key changes (e.g., a `CustomerID` updates), all linked transactional historical entries safely update instantly across dependent tables.
- **ON DELETE RESTRICT:** To protect transaction ledger history, a primary record (like a `Customer` or `Product`) cannot be deleted if active historic dependencies point to it (such as matching active `Order` or `OrderItem` records).